Identifying Information:

NAME: Beck, Marcus W.
PERSISTENT IDENTIFIER (PID): https://orcid.org/0000-0002-4996-0059
POSITION TITLE: Senior Scientist
PRIMARY ORGANIZATION AND LOCATION: Tampa Bay Estuary Program, St. Petersburg, Florida, United States

Professional Preparation:

ORGANIZATION AND LOCATION	DEGREE (if applicable)	RECEIPT DATE	FIELD OF STUDY
University of Minnesota, Twin Cities, Minnesota, United States	Ph.D.	08/2009 - 06/2013	Conservation Biology (Fisheries and Aquatic Biology); Statistics minor
University of Minnesota, Twin Cities, Minnesota, United States	M.Sc.	08/2007 - 06/2009	Conservation Biology (Fisheries and Aquatic Biology)
University of Florida, Gainesville, Florida, United States	B.Sc.	08/2004 - 12/2006	Zoology
Santa Fe Community College, Gainesville, Florida, United States	A.A.	08/2002 - 04/2004	Zoology

Appointments and Positions

2019 - present	Senior Scientist, Tampa Bay Estuary Program, St. Petersburg, Florida, United States
2021 - present	Private Contractor, EnviroDev, LLC, United States
2017 - 2019	Scientist, Southern California Coastal Water Research Project, Costa Mesa, California, United States
2015 - 2017	Post-Doctorate Researcher, Ecosystem Dynamics and Effects Branch, Gulf Ecology Division, United States Environmental Protection Agency, Gulf Breeze, Florida, United States
2013 - 2015	Post-Doctorate Research Fellow, Oak Ridge Institute for Science and Education, Ecosystem Dynamics and Effects Branch, Gulf Ecology Division, United States Environmental Protection Agency, Gulf Breeze, Florida, United States

Products

1. Yang B, Liu Y, Law JA, Weisberg RH, D'Angelo S, Qiao K, Beck MW, Sherwood ET, Frazer TK. Tampa Bay Observing Network (TBON): A continuous data acquisition system in support of science and environmental management. Environmental Monitoring and Assessment. 2026; 198:1149. Available from: https://doi.org/10.1007/s10661-026-16007-4

2. Beck MW, Hagy JD, Le C. Quantifying seagrass light requirements using an algorithm to spatially resolve depth of colonization. Estuaries and Coasts. 2018; 41(2):592-610. Available from: https://doi.org/10.1007/s12237-017-0287-1

3. Beck MW, Flaherty-Walia KE, Scolaro S, Burke MC, Furman BT, Karlen DJ, Pratt C, Anastasiou CJ, Sherwood ET. Hot and fresh: Evidence of climate-related suboptimal water conditions for seagrass in a large Gulf coast estuary. Estuaries and Coasts. 2024; 47:1475-1497. Available from: https://doi.org/10.1007/s12237-024-01385-0

4. Le C, Lehrter JC, Schaeffer BA, Hu C, Murrell MC, Hagy JD, Greene RM, Beck MW. Bio-optical water quality dynamics observed from MERIS in Pensacola Bay, Florida. Estuarine, Coastal and Shelf Science. 2016; 173:26-38. Available from: https://doi.org/10.1016/j.ecss.2016.02.003

5. Lizcano-Sandoval L, Beck MW, Scolaro S, Sherwood ET, Muller-Karger FE. Cloud-based satellite remote sensing for enhancing seagrass monitoring and ecosystem management. ISPRS Journal of Photogrammetry and Remote Sensing. 2025; 227:508-518. Available from: https://doi.org/10.1016/j.isprsjprs.2025.06.034

6. Beck MW, Sherwood ET, Murphy RR, Flaherty-Walia KE, Zhang Q, Simmons BA, Burke MC. Managing the unmanageable: Understanding the competing effects of salinity and nutrient loading on estuarine water quality. Estuaries and Coasts. 2026; 49:178. Available from: https://doi.org/10.1007/s12237-026-01792-5

7. Arriola JM, Najjar RG, Briceño H, Hu C, Herrmann M, Beck MW. Ecosystem metabolic rates estimated from diel oxygen measurements in two subtropical estuaries. Estuaries and Coasts. 2025; 48:155. Available from: https://doi.org/10.1007/s12237-025-01597-y

8. Medina M, Beck MW, Hecker J, Iadevaia N, Moody B, Anastasiou C, Tomasko D, Milbrandt EC, Kaplan D, Angelini C. Water quality trends and eutrophication indicators in a large subtropical estuary: A case study of the Greater Charlotte Harbor system in southwest Florida. Estuaries and Coasts. 2025; 48:56. Available from: https://doi.org/10.1007/s12237-025-01488-2

9. Beck MW, Arriola JM, Herrmann M, Najjar RG. Fitting metabolic models to dissolved oxygen data: The Estuarine BAyesian Single-station Estimation method (EBASE). Limnology and Oceanography: Methods. 2024; 22(8):590-607. Available from: https://doi.org/10.1002/lom3.10620

10. Beck MW, de Valpine P, Murphy R, Wren I, Chelsky A, Foley M, Senn DB. Multi-scale trend analysis of water quality using error propagation of generalized additive models. Science of the Total Environment. 2022; 802:149927. Available from: https://doi.org/10.1016/j.scitotenv.2021.149927

11. Beck MW, Altieri A, Angelini C, Burke MC, Chen J, Chin DW, Gardiner J, Hu C, Hubbard KA, Liu Y, Lopez C, Medina M, Morrison E, Phlips EJ, Raulerson GE, Scolaro S, Sherwood ET, Tomasko D, Weisberg R, Whalen J. Initial estuarine response to inorganic nutrient inputs from a legacy mining facility adjacent to Tampa Bay, Florida. Marine Pollution Bulletin. 2022; 178:113598. Available from: https://doi.org/10.1016/j.marpolbul.2022.113598

12. Beck MW, Burke MC, Raulerson GE, Scolaro S, Sherwood ET, Whalen J. Coordinated monitoring of the Piney Point wastewater discharge into Tampa Bay: Data synthesis and reporting. Florida Scientist. 2023; 86(2):288-300.

13. Beck MW, Wetherill B, Carr J. MassWateR: Improving quality control, analysis, and sharing of water quality data. PLOS ONE. 2023; 18(11):e0293737. Available from: https://doi.org/10.1371/journal.pone.0293737

14. Le C, Lehrter JC, Hu C, MacIntyre H, Beck MW. Satellite observation of particulate organic carbon dynamics on the Louisiana continental shelf. Journal of Geophysical Research: Oceans. 2017; 122(1):555-569. Available from: https://doi.org/10.1002/2016JC012275

15. Beck MW, Murphy R. Numerical and qualitative contrasts of two statistical models for water quality change in tidal waters. Journal of the American Water Resources Association. 2017; 53(1):197-219. Available from: https://doi.org/10.1111/1752-1688.12489

16. Beck MW, Schrandt MN, Wessel MR, Sherwood ET, Raulerson GE, Prasad AAB, Best BD. tbeptools: An R package for synthesizing estuarine data for environmental research. Journal of Open Source Software. 2021; 6(65):3485. Available from: https://doi.org/10.21105/joss.03485

17. Beck MW, de Valpine P, Murphy R, Wren I, Chelsky A, Foley M, Senn DB. wqtrends: Assess water quality trends with generalized additive models, v1.5.2. CRAN R package. 2026. Available from: https://tbep-tech.github.io/wqtrends/

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19. Le C, Wu S, Hu C, Beck MW, Yang X. Phytoplankton decline in the eastern North Pacific transition zone associated with atmospheric blocking. Global Change Biology. 2019; 25(10):3485-3493. Available from: https://doi.org/10.1111/gcb.14737

20. Le C, Wu M, Sun H, Long S, Beck MW. Linking phytoplankton variability to atmospheric blocking in an eastern boundary upwelling system. Journal of Geophysical Research: Oceans. 2022; 127(6):e2021JC017348. Available from: https://doi.org/10.1029/2021JC017348

Certification:

I certify that the information provided is current, accurate, and complete. This includes but is not limited to information related to domestic and foreign appointments and positions.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

In accordance with section 10634 of the CHIPS and Science Act of 2022 (42 U.S.C. § 19234), I certify that I have completed the requisite research security training that meets the requirements specified in NASA's research security training requirements guidance within 12 months prior to proposal submission.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Certified by Beck, Marcus on 2026-10-08 18:02:31